

4. Project Planning Phase

Project: PocketSmart AI - Smart Budget & Recommendation Assistant

4.1 Introduction

The Project Planning Phase defines how the PocketSmart AI project will be organized and developed. Based on the project workflow, the work is divided into major activities covering Gemini AI setup, core functionality development, backend implementation, user-interface development, and testing and optimization. This planned structure provides a clear path from the initial idea to a working application.

4.2 Planning Objectives

The main objective is to complete the project in a systematic way by separating the work into manageable milestones. Each milestone has a specific purpose and a set of activities. The planning also helps ensure that the Home, Party, and Jewelry planners, along with authentication and recommendation features, are included in the implementation.

4.3 Major Work Areas

The planned work areas are AI initialization, core functionality development, FastAPI backend integration, frontend and UI development, and testing and optimization. These areas correspond to the documented project workflow and milestones.

4.4 Development Approach

The project uses a modular development approach. Backend routes, services, and models are organized separately, while planner-specific logic is connected to a common Gemini AI recommendation layer. This approach allows the different project components to be developed and tested in an organized sequence.

4.5 Planning Outcome

The result of the planning phase is a structured development roadmap. The roadmap identifies the main technical tasks, required resources, expected outputs, testing activities, and areas where validation or fallback handling must be included.

Page 2 - Work Breakdown and Milestones

4.6 Milestone 1 - Gemini AI Initialization

The first planned milestone is the setup of the AI infrastructure. The documented activities include creating or configuring Google Cloud access, enabling Gemini API access, configuring API keys and permissions, setting usage limits, and validating connectivity. Sample text-only and image-plus-text prompts are planned to confirm multimodal compatibility.

4.7 Milestone 2 - Core Functionalities Development

The second milestone focuses on the core backend intelligence. Planned activities include designing the backend structure, creating endpoints for Home Planner, Party Planner, and Jewelry Planner, and integrating planner-specific Gemini prompts and templates. The workflow also includes optional fallback strategies when suitable product matches are limited.

4.8 Milestone 3 - Backend FastAPI Integration

The third milestone focuses on implementing the FastAPI-powered backend. Planned activities include defining API routes, setting up modular architecture, configuring CORS and static routing, supporting authentication and session functions, and preparing startup and application entry behavior.

4.9 Milestone 4 - UI Development

The fourth milestone focuses on the user interface. The plan includes creating responsive HTML forms for Home Interior, Party, and Jewelry planners, connecting them to FastAPI POST routes, and displaying AI responses in structured card-like layouts.

4.10 Milestone 5 - Testing and Optimization

The fifth milestone focuses on validating the reliability and contextual awareness of the recommendation engine. The plan includes real-world budget testing, evaluation of Gemini response quality, prompt optimization, input validation, and fallback or default recommendations when needed.

Page 3 - Task and Resource Planning

4.11 AI and API Resources

The project requires access to the Gemini API and the associated authentication or API-key configuration. The documented workflow also identifies Python and FastAPI as development technologies and requires configuration of environment variables for application settings.

4.12 Software Development Resources

The planned software resources include Python for backend development, FastAPI for request handling and routing, HTML/CSS and JavaScript for the frontend, Jinja2 templates for page rendering, and Uvicorn for running the FastAPI application. These technologies are used together to implement the documented architecture.

4.13 Backend Task Planning

Backend tasks are planned around modular routes, services, and models. Route development covers the planner endpoints and authentication endpoints. Service development includes Gemini prompt orchestration, recommendation logic, budget formatting, domain segmentation, and image analysis where required.

4.14 Frontend Task Planning

Frontend tasks include designing the Home Page, Register and Login pages, User Dashboard, planner forms, recommendation pages, Testimonials Page, Footer, and Recommendation History Page. Dynamic JavaScript behavior is planned for displaying recommendation results clearly.

4.15 Data Source Planning

The project design refers to external platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, and OYO for product and service information. Where direct APIs are unavailable during development, the workflow allows mock API calls or simulated sourcing for testing and implementation.

4.16 Task Dependencies

The planned sequence creates practical dependencies. AI access and connectivity need to be established before AI-based recommendation functions can be tested. Backend routes and services need to be available before the frontend forms can communicate with them. End-to-end testing follows after the major modules are integrated.

Page 4 - Schedule, Testing, and Risk Planning

4.17 Development Sequence

The project workflow provides a sequence from AI initialization to core functionality, backend integration, UI development, and finally testing and optimization. This sequence allows the development team to establish the AI foundation first and then build the application layers around it.

4.18 Testing Plan

Testing is planned across Home Interior, Party, and Jewelry scenarios using varied budgets and preferences. Home tests include decor setups using sources such as IKEA and Amazon. Party tests include food, venue, and decoration packages. Jewelry tests include occasion, outfit style, and image inputs.

4.19 AI Response Evaluation

The project plan includes evaluating the quality of Gemini responses, budget adherence, and platform accuracy. The results of testing can be used to refine the prompt structures so that recommendations better match the user's stated requirements.

4.20 Input Validation Plan

Input validation is included in the optimization plan to prevent failures caused by edge cases. The system should check important user information before processing a request and should handle cases where AI output does not contain sufficient recommendations.

4.21 Fallback Planning

Fallback or default recommendations are planned as a support mechanism when the AI returns insufficient results. This helps the application continue to provide useful output instead of failing completely.

4.22 Authentication and Session Planning

Authentication tasks include registration, login, logout, token generation, session information, and session data. These functions are planned to support personalized interactions, protected API routes, and recommendation tracking.

4.23 Maintenance Planning

The modular architecture supports future maintenance because routes, services, models, and planner functions are separated. Prompt templates and recommendation logic can therefore be refined without redesigning the entire application.

Page 5 - Deliverables and Final Project Plan

4.24 Planned Project Deliverables

The planned deliverables include a working FastAPI backend, Gemini AI integration, Home Interior Planner, Party Planner, Jewelry Planner, authentication functions, session handling, frontend pages, recommendation displays, and recommendation history. These deliverables correspond to the features and workflow described in the project documentation.

4.25 Backend Deliverables

The backend deliverables include planner routes such as /generate-home, /generate-party, and /generate-jewelry; authentication routes such as /register, /login, /logout, and /token; session routes; recommendation detail handling; history retrieval; startup initialization; and the modular services and models needed for the application.

4.26 Frontend Deliverables

The frontend deliverables include the Home Page, Testimonials Page, Register Page, Login Page, User Dashboard, planner pages, recommendation pages, Recommendation History Page, and supporting interface elements. The UI is planned to be responsive and to display AI outputs in structured layouts.

4.27 Success Criteria

The project plan considers the system ready for evaluation when the main planner workflows operate successfully, user inputs are handled correctly, Gemini connectivity works as intended, recommendations are displayed clearly, authentication and session functions operate, and testing has been performed across the three planning domains.

4.28 Final Project Plan

The complete plan moves through five major stages: establish Gemini AI access, develop core planner functionality, integrate the FastAPI backend, build the responsive user interface, and perform testing and optimization. Each stage has defined activities and outputs that contribute to the final PocketSmart AI application.

4.29 Conclusion

The Project Planning Phase organizes the development of PocketSmart AI into clear milestones and tasks. It identifies the resources required, the order of development, the major deliverables, and the testing activities needed to validate the application. This structured plan provides a practical roadmap for moving from project design to implementation and final evaluation.